# Aqua-Sense: Ai-Powered Iot System For Public Water Quality Monitoring & Predictive Alert

## LITERATURE REVIEW

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## 1. Introduction

Water quality monitoring is essential for ensuring safe and reliable water supplies and supporting effective environmental and public-health management. Traditional water-quality assessment often relies on periodic sampling and laboratory analysis, which may not provide continuous visibility into changing water conditions.

Recent research has explored the use of Internet of Things (IoT) technologies, wireless communication, cloud platforms, and machine learning to enable real-time and remote water-quality monitoring. The reviewed studies cover IoT-based sensing, long-range communication, cloud-based monitoring, machine-learning-based prediction, mobile applications, and intelligent data analysis.

This literature review examines existing approaches to identify their major contributions, limitations, and research gaps relevant to the development of AquaSense.

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## Paper 1 – IoT-Enabled Advanced Water Quality Monitoring System for Pond Management and Environmental Conservation

**Authors:** Jeetendra Kumar, Rashmi Gupta, Suvarna Sharma, Tulika Chakrabarti, Prasun Chakrabarti, Martin Margala
**Year:** 2024

**Methodology:**
The study uses an ESP32 microcontroller with pH, turbidity, and TDS sensors. The sensors are mounted on an aquatic boat to collect water-quality data from different areas of a pond.

The collected data is uploaded to the cloud and visualized through the AquaSpecs application.

**Key Contributions:**

- Enables real-time monitoring at multiple pond locations.
- Provides cloud-based analysis and mobile accessibility.
- Reports lower sensor error compared with traditional laboratory methods.

**Limitations / Gaps:**

- Monitoring is limited to three parameters: pH, TDS, and turbidity.
- Scalability and long-term maintenance were not fully explored.

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## Paper 2 – A Low-Cost LoRa Optical Fluorometer Nephelometer for Wireless Monitoring of Water Quality Parameters in Real Time

**Authors:** Soheyl Faghir Hagh, Parmida Amngostar, Wenzhe Chen, Jessica Sheridan, Clayton J. Williams, Ana M. Morales-Williams, Dryver Huston, Tian Xia
**Year:** 2024

**Methodology:**
The study presents a portable IoT-enabled device using optical sensors to measure turbidity, temperature, chlorophyll-a, and phycocyanin. LoRaWAN is used for wireless communication, while an SD card provides local data backup.

**Key Contributions:**

- Provides an affordable and customizable monitoring solution.
- Uses low-power communication suitable for field deployment.
- Includes biological indicators such as algae and cyanobacteria.

**Limitations / Gaps:**

- Validation under real-world outdoor conditions was limited.
- The number of sensing parameters is narrower than that of commercial water-quality devices.

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## Paper 3 – Smart System for Water Quality Monitoring Utilizing Long-Range IoT

**Authors:** Muhammad Ary Murti, Andi Rudi Adhy Saputra, Ibnu Alinursafa, Ali Najah Ahmed, Ayman Yafouz, Ahmed El-Shafie
**Year:** 2024

**Methodology:**
The system uses an ATmega328P-AU microcontroller with pH and turbidity sensors. Data is transmitted through LoRa, using LPWAN communication, to the Antares cloud service. Results are displayed through an Android application.

**Key Contributions:**

- Provides low-cost, real-time water-quality monitoring.
- Uses long-range communication with relatively low power consumption.
- Reports pH and turbidity errors of 0.27% and 0.59%, respectively.

**Limitations / Gaps:**

- Only pH and turbidity are monitored.
- Sensor calibration and periodic maintenance are required.
- Operation is limited by LoRa coverage.

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## Paper 4 – A Hybrid Machine Learning and Embedded IoT-Based Water Quality Monitoring System

**Authors:** Ismail A. Adeleke, Nnamdi I. Nwulu, Omolola A. Ogbolumani
**Year:** 2022

**Methodology:**
The system integrates multiple water-quality sensors, including pH, dissolved oxygen, TDS, electrical conductivity, ORP, turbidity, and temperature, with Arduino and ESP8266 for data acquisition. Data is transmitted to ThingSpeak and processed using Artificial Neural Network (ANN) and Support Vector Machine (SVM) models on Google Cloud. An automated control unit can activate corrective water treatment based on threshold values.

**Key Contributions:**

- Supports real-time and remote water-quality monitoring.
- Applies machine learning for water-quality prediction.
- Provides automated corrective water-treatment functionality.
- Reports ANN performing better than SVM in prediction.

**Limitations / Gaps:**

- Requires extensive training data and computational resources.
- Neural networks have limited extrapolation capabilities.
- Performance depends on sensor calibration and stable cloud connectivity.

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## Paper 5 – Toward Design of IoT and Machine Learning-Enabled Frameworks for Analysis and Prediction of Water Quality

**Authors:** Mushtaque Ahmed Rahu, Abdul Fattah Chandio, Khursheed Aurangzeb, Sarang Karim, Musaed Alhussein, Muhammad Shahid Anwar
**Year:** 2023

**Methodology:**
The framework uses IoT sensors including pH, temperature, turbidity, and TDS. The collected data is preprocessed and analyzed using machine-learning models. Regression models such as MLP and SVR are used for Water Quality Index (WQI), while SVM and Random Forest are used for water-quality classification.

**Key Contributions:**

- Supports continuous and real-time water-quality monitoring.
- Combines IoT sensing with machine-learning analysis.
- Enables early identification of potential water-quality problems.
- Supports decision-making for environmental and public-health management.

**Limitations / Gaps:**

- Lower performance is reported on large datasets.
- Performance depends on sensor accuracy and connectivity.
- Implementation and maintenance can be costly.

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## Paper 6 – IoT Real-Time Potable Water Quality Monitoring and Prediction

**Authors:** Rita Wiryasaputra, Chin-Yin Huang, Yu-Ju Lin, Chao-Tung Yang
**Year:** 2024

**Methodology:**
The system uses TDS, pH, dissolved oxygen, and temperature sensors with an Arduino-based data-acquisition system. NB-IoT is used for communication, while cloud monitoring is provided through Grafana and notifications through the LINE application. Several machine-learning models, including Decision Tree, Random Forest, Gradient Boosting, SVM, and Neural Network, are applied for prediction.

**Key Contributions:**

- Provides multi-parameter water-quality monitoring.
- Combines real-time monitoring with predictive analysis.

- Provides web/mobile monitoring and alerts.
- The study reports more than 97% accuracy for Decision Tree and SVM models.

**Limitations / Gaps:**

- The system has greater complexity and cost than basic IoT systems.
- Machine-learning models require sufficient training data.
- The system depends on stable NB-IoT/5G connectivity.

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## Paper 7 – IoT-Based Water Quality Monitoring in Philippine Off-Grid Communities

**Authors:** Jenny Vi Abrajano, Jenalyn Apanay, Khavee Agustus Botangen, Jovith Nabua, Chezalea Fay Peña
**Year:** 2024

**Methodology:**
The system uses an Arduino-based platform with pH, turbidity, and temperature sensors. Data is transmitted to Google Firebase through Wi-Fi, while GSM SMS is used for alerts. The system also includes solar power with battery backup and web/mobile applications for monitoring.

**Key Contributions:**

- Designed for low-cost monitoring in rural and off-grid communities.
- Provides real-time monitoring and SMS alerts.
- Uses solar power and battery backup.
- Provides accessible web and mobile applications for community use.

**Limitations / Gaps:**

- Full functionality depends on internet connectivity.
- Only a limited number of water-quality parameters are monitored.
- Sensors require appropriate maintenance and calibration.

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## Paper 8 – Wireless Sensor Network IoT Platform for Water Consumption and Quality

**Authors:** Christodoulos Axiotidis, Evangelia Konstantopoulou, Nicolas Sklavos
**Year:** 2024

**Methodology:**
The study presents an ESP32-based IoT platform using turbidity, pH, temperature, ORP, and flow sensors. Long-range wireless communication is used to transmit data to Firebase. Three applications are developed for consumers, technicians, and administrators.

**Key Contributions:**

- Monitors water consumption and quality simultaneously.
- Uses long-range wireless communication.
- Integrates Firebase for data handling.
- Provides separate applications for different user groups.
- Provides a cost-effective alternative to traditional laboratory testing.

**Limitations / Gaps:**

- Does not include some important parameters such as nitrates, chlorine, and microbial indicators.
- Requires regular sensor calibration.
- The prototype was tested under controlled conditions, requiring further validation for real-world scalability.

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## Paper 9 – Connected Sensors, Innovative Sensor Deployment, and Intelligent Data Analysis for Online Water Quality Monitoring

**Authors:** Siti Nadhirah Zainurin, Wan Zakiah Wan Ismail, Siti Nurul Iman Mahamud, Irneza Ismail, Juliza Jamaludin, Khairul Nabilah Zainul Ariff, Wan Maryam Wan Ahmad Kamil
**Year:** 2022

**Methodology:**
The study integrates electrochemical, optical, and biosensors into connected networks. Sensors are deployed using platforms such as buoys, robots, and drones to provide wider spatial and temporal coverage. Artificial intelligence is applied for standardized data analysis and forecasting.

**Key Contributions:**

- Supports continuous and real-time monitoring.
- Provides broader spatial and temporal coverage than conventional laboratory methods.
- Uses AI for data analysis and forecasting.
- Supports early detection of contaminants and decision-making.

**Limitations / Gaps:**

- Sensor drift and fouling can affect long-term stability.
- Deployment and maintenance can be expensive.
- Connectivity, power, and large-scale data handling remain challenges.

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## Paper 10 – AQUASENSE – Sensor-Based Water Quality Monitoring Device

**Authors:** S. Hema Priyadarshini, Vinay K. V., Poojitha S., Ashrita Dikshit V A, Rakshit Preran B.
**Year:** 2023

**Methodology:**
The system uses an Arduino UNO with TDS, pH, and temperature sensors. Water samples

from different locations are measured and the readings are displayed on an LCD. The results are compared with Indian water-quality standards.

**Key Contributions:**

- Provides a low-cost and portable monitoring system.
- Provides real-time results for basic water-quality parameters.
- Offers an alternative to commercial water-quality monitoring devices.

**Limitations / Gaps:**

- Limited to physical parameters such as TDS, pH, and temperature.
- Accuracy can vary due to sensor limitations and environmental conditions.
- Cannot detect chemical and biological contaminants.

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### 3. Comparative Analysis

| Study | Main Technology / Method | Main Parameters / Focus | Major Contribution | Major Limitation |
|---|---|---|---|---|
| IoT-Enabled Advanced Water Quality Monitoring System | ESP32 + sensors + cloud + mobile app | pH, turbidity, TDS | Real-time pond monitoring | Limited parameters; scalability not fully explored |
| Low-Cost LoRa Optical Fluorometer | Optical sensors + LoRaWAN + SD backup | Turbidity, temperature, chlorophyll-a, phycocyanin | Low-cost, low-power field monitoring | Limited outdoor validation |
| Hybrid ML and Embedded IoT System | Arduino/ESP8266 + cloud + ANN/SVM | Multiple water-quality parameters | Predictive analysis and automated treatment | Training data and computational requirements |

| Study | Main Technology / Method | Main Parameters / Focus | Major Contribution | Major Limitation |
|---|---|---|---|---|
| Connected Sensors and Intelligent Data Analysis | Connected sensors + AI | Multi-parameter monitoring | Continuous monitoring and forecasting | Sensor drift, cost, connectivity |
| IoT and ML-Enabled Framework | IoT + MLP/SVR/SVM/RF | pH, temperature, turbidity, TDS | WQI and water-quality classification | Dataset and sensor limitations |
| Smart Long-Range IoT System | LoRa + cloud + Android app | pH, turbidity | Long-range, low-power monitoring | Only two parameters |
| IoT Real-Time Potable Water Monitoring | NB-IoT + cloud + ML | TDS, pH, DO, temperature | Prediction and real-time alerts | Complexity, cost, connectivity |
| AQUASENSE Sensor-Based Device | Arduino + sensors + LCD | TDS, pH, temperature | Low-cost portable monitoring | Limited detection capability |
| Philippine Off-Grid Monitoring | Arduino + Firebase + GSM + solar | pH, turbidity, temperature | Rural/off-grid monitoring and alerts | Internet and parameter limitations |
| Wireless Sensor Network Platform | ESP32 + LoRa + Firebase + multiple apps | Quality + consumption | Multi-user monitoring and long-range communication | Limited parameters and prototype validation |

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## 4. Research Gaps

The reviewed studies demonstrate that IoT, wireless communication, cloud platforms, and machine learning can significantly support water-quality monitoring. However, several limitations can be observed across the reviewed approaches.

### 4.1 Limited Multi-Parameter Monitoring

Several reviewed systems monitor only a small number of parameters, such as pH, turbidity, TDS, or temperature. Some studies include additional parameters, but the range differs between systems.

### 4.2 Limited Integration of Monitoring and Prediction

Many systems focus primarily on real-time sensing and visualization, while predictive analysis is addressed separately in other studies. Integrating continuous sensing with predictive analysis can provide a more comprehensive monitoring approach.

### 4.3 Connectivity and Cloud Dependency

Existing systems use different communication technologies, including Wi-Fi, LoRa, LoRaWAN, NB-IoT, and GSM. Connectivity availability and reliability remain important considerations, particularly for remote or distributed monitoring environments.

### 4.4 Sensor Calibration and Long-Term Reliability

Sensor calibration, sensor drift, fouling, environmental conditions, and maintenance are recurring limitations identified across the reviewed studies. These factors can affect the reliability of long-term water-quality monitoring.

### 4.5 Limited Explainability of AI-Based Predictions

Machine-learning-based studies demonstrate the potential for prediction and classification, but the reviewed literature does not consistently address how the factors contributing to an AI prediction can be communicated to end users.

### 4.6 Real-World Scalability

Some systems are tested as prototypes or under controlled conditions. Long-term deployment, maintenance, scalability, and operation under real-world environmental conditions require further consideration.

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## 5. Relevance to AQUA-SENSE

The reviewed studies provide the technological and research foundation for AquaSense. The literature demonstrates the feasibility of combining IoT-based sensing, wireless

communication, cloud platforms, mobile applications, and machine learning for water-quality monitoring.

Based on the reviewed approaches, AquaSense brings together several relevant concepts within a single proposed system:

- **IoT-based sensing** for collecting water-quality measurements.
- **ESP32-based sensor integration** for acquiring multi-parameter readings.
- **LoRa communication** for wireless transmission between the sensor node and base station.
- **Firebase Realtime Database** for cloud-based storage and access to sensor readings.
- **AI-based analysis** using a hybrid CNN-LSTM approach for water-quality classification and analysis.
- **Mobile application** for displaying monitoring information and alerts.
- **Explainable AI (XAI)** to support recommendations and provide greater interpretability of AI-based results.
- **User feedback and authority interaction** to support a more responsive water-quality management approach.

The reviewed literature therefore supports the overall direction of AquaSense while also highlighting practical challenges such as sensor calibration, connectivity, training-data requirements, model performance, and real-world scalability.

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## 6. Conclusion

The literature review shows that IoT-based water-quality monitoring has developed from basic sensor measurement systems toward more connected platforms incorporating cloud services, long-range communication, mobile applications, and machine learning.

Existing studies demonstrate the benefits of real-time monitoring, remote access, predictive analysis, and automated alerts. At the same time, recurring challenges include limited sensing parameters, sensor calibration, connectivity dependence, training-data requirements, deployment cost, and limited real-world validation.

AquaSense is developed in this research context by combining IoT sensing, LoRa communication, Firebase-based cloud data handling, AI-based analysis, and a mobile application. The reviewed studies provide the technical background for these components and help identify areas that require further consideration during implementation and validation.

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## 7. References


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